

Package ‘skrmdb’

October 8, 2026

Type Package

Title Package to estimate ED50

Version 5.0.5

Date 2025-10-08

Description Package to estimate ED50 by the methods of Spearman-Kärber, Reed-Muench, Dragstedt-Behrens. No endorsement, claim, or warranty is implied for this package. It is made available for investigational or pedagogical use only.

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URL <https://aapiaame0vgit10.usda.net/statistics/skrmdb/>

BugReports <https://aapiaame0vgit10.usda.net/statistics/skrmdb/-/issues/>

News NEWS.md

LazyData true

Depends R (>= 4.3)

Imports CVBformula (>= 0.0.0.9008),
crayon,
cli,
data.table,
lifecycle,
stringr

Suggests spelling,
testthat,
knitr,
rmarkdown,
R.rsp

Encoding UTF-8

VignetteBuilder R.rsp

Config/testthat/edition 3

Config/roxygen2/markdown TRUE

Config/roxygen2/version 8.1.0

Language en-US

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DragBehr

Functions for computing ED50

Description

The skrmdb package provides functionality to compute the median effective dose (ED50) using the Dragstedt-Behrens, Reed-Muench, and Spearman-Kärber estimators.

The Dragstedt-Behrens and Reed-Muench methods estimate the median effective dose by interpolating between the two doses that bracket the dose producing median response. They accumulate sums in both directions by assuming that subjects that responded at a lower dose would respond at a higher dose, and subjects that did not respond at a higher dose would not respond at a lower dose. The Dragstedt-Behrens method estimates ED50 by interpolating on the line that connects the hypothetical fractions of the bracketing doses for ED50, while the Reed-Muench method estimates ED50 as the intersection of the lines connecting the two sets of cumulative sums between bracketing doses.

The Spearman-Kärber method gives a non-parametric estimate of the mean of a tolerance distribution from its empirical distribution (EDF). The empirical PMF is derived from the EDF by differencing. The estimator is $\sum x f(x)$. If the EDF does not cover the entire support of x , SpearKarb() extends it by assuming the next lower dilution would produce zero response and the next higher dilution would produce complete response.

The function `skrmdb.all()` reports the results for all three methods.

There are several assumptions that these methods make:

1. The group size n_i is constant.
 - Group sizes and responses are scaled silently if this condition is not met.
2. The log dilutions $\mathbf{x} = \{x_1, x_2, \dots, x_k\}$ form an increasing sequence.
 - The data will be automatically sorted so this condition holds. If a dilution x appears more than once in $\mathbf{x}$, a new dilution sequence will be created where each dilution appears once. For each dilution x , the corresponding n_i values are summed to create a new n sequence, and the corresponding y_i values are summed to create a new y sequence. The column **Duplicate** in the output reports "Yes" if a dilution appears more than once in the original sequence and "No" otherwise.
3. That $\mathbf{x}$ is an arithmetic sequence. That is, there is some number Δ such that $x_k = x_1 + (k - 1) \times \Delta$ for all k .
 - The column **Dilutions** in the output reports "regular" if this condition is met and "irregular" otherwise. Generally, this is used to indicate the possibility of a missing dilution in a sequence. However, it is possible that in some cases, a sequence which is marked as "irregular" may in fact be regular and have no missing dilutions. In these cases, use your best judgment.

4. The count ratio sequence $\mathbf{r} = \{r_1, r_2, \dots, r_k\}$ defined by $r_i = y_i/n_i$ trends from smaller to larger.
 - The column **Response** indicates whether $\mathbf{r}$ is "increasing", "decreasing", or shows "no change". If it is "decreasing" and autosort is TRUE, then ED50 will be computed using the sequence $1 - r_i$, which is the appropriate transformation for decreasing response data. When the data is very noisy and the trend is not clear, it is likely that using these methods to compute ED50 is not appropriate.
5. Preferably, the count ratio sequence is monotonic, that is $r_i \leq r_{i+1}$ for all i or $r_i \geq r_{i+1}$ for all i .
 - The column **Monotonic** indicates if this condition holds. Real-world data is rarely monotonic, but good data is generally "close enough to monotonic for all practical purposes." Once again, use your best judgment.
6. The log dilution sequence brackets ED50. That is, there is an i such that $x_i \leq \text{ED50} \leq x_{i+1}$. Note: This is **not** equivalent to saying that there is an i such that $r_i \leq 0.5 \leq r_{i+1}$. However, if it is the case that $0.5 < r_1$ or $r_k < 0.5$, then the log dilution sequence will not bracket ED50.
 - The column **Bracket** indicates if this condition holds. If it does not hold, use your best judgment as to whether or not ED50 could be considered to be meaningful.

Usage

```
DragBehr(
  data,
  formula,
  autosort = TRUE,
  warn.me = TRUE,
  show = FALSE,
  y = deprecated(),
  n = deprecated(),
  x = deprecated()
)
```

```
ReedMuench(
  data,
  formula,
  autosort = TRUE,
  warn.me = TRUE,
  show = FALSE,
  y = deprecated(),
  n = deprecated(),
  x = deprecated()
)
```

```
skrmdb.all(data, formula, autosort = TRUE, warn.me = TRUE)
```

```
SpearKarb(
  data,
  formula,
  autosort = TRUE,
  warn.me = TRUE,
  show = FALSE,
  y = deprecated(),
  n = deprecated(),
  x = deprecated()
)
```

```

    n = deprecated(),
    x = deprecated()
  )

```

Arguments

data	A <code>data.frame</code> containing the titration data. Formatted as specified in the CVB Data Guide.
formula	A formula of the form $y + n \sim x$ or $y + n \sim x \mid w_1 + \dots + w_n$ where $w_1 \dots w_n$ are grouping variables. All variables must be distinct.
autosort	If TRUE, the functions will compute the ED50 based on either y / n or $1 - y / n$, whichever appears to increase with x . This is how the three methods assume the data to be ordered. If FALSE, ED50 will be computed using y / n , which could give incorrect results. Do not change this parameter unless you are certain you know what you are doing.
warn.me	If TRUE, warnings and messages related to the processing of the data will be displayed. These warnings correspond to the assumptions and feedback columns in the output as outlined in the Description section above.
show	If TRUE, will print the intermediate statistics used to calculate ED50. These statistics are captured in the data component of the output.
y	[Deprecated] An integer vector corresponding to the number responding at each log dilution or dose.
n	[Deprecated] An integer vector corresponding to the group size at each log dilution or dose.
x	[Deprecated] A vector corresponding to the log dilution or dose for each group.

Value

A list of class `skrmdb` which contains the following elements

- `eval`: Which method or methods were used to compute ED50.
- `data`: The transformed data used to compute ED50. It contains the columns `y`, `n`, `x`, and any extra columns on which the results are conditioned. The columns `y_inc` and `y_dec` contain the increasing and decreasing versions of the response proportions, based on y / n and $1 - y / n$. The columns `Duplicate`, `Dilutions`, `Response`, `Monotonic`, and `Bracket` give information about the how well the data for each group meets each of the assumptions and can be used to find specific parts of the data which do not meet a particular assumption.
- `ed`: ED50. If returned from `skrmdb.all()`, then NA.
- `var`: The variance. Only meaningful for `SpearKarb()`.
- `results`: A `data.frame` giving the ED50 for each group based on the conditions. it also contains the columns `Duplicate`, `Dilutions`, `Response`, `Monotonic`, and `Bracket` which describe how well the data for that group meets each of the assumptions.
- `autosort`: The value of the parameter `autosort`.

Note

Many microbiology texts mistakenly present the Dragstedt-Behrens method as the Reed-Muench method.

References

- Behrens, B. (1929) Zur Auswertung der Digitalisblätter im Froschversuch. *Arkiv für Experimentelle Pathologie und Pharmakologie*. **140**: 237-256.
- Dragstedt, C. A., Lang, V. F. (1928). Respiratory Stimulants in Acute Cocaine Poisoning in Rabbits. *J. of Pharmacology and Experimental Therapeutics*. **32**: 215-222.
- Kärber, G. (1931). Beitrag zur kollektiven Behandlung Pharmakologischer Reihenversuche. *Archiv für Experimentelle Pathologie und Pharmakologie*. **162**: 480-483.
- Miller, Rupert G. (1973). Nonparametric Estimators of the Mean Tolerance in Bioassay. *Biometrika*. **60**: 535 - 542.
- Reed, L.J., Muench, H. (1938). A Simple Method of Estimating Fifty Percent Endpoints. *American Journal of Hygiene*. **27**: 493-497.
- Spearman, C. (1908). The Method of "Right and Wrong Cases" ("Constant Stimuli") without Gauss's Formulae. *Brit. J. of Psychology*. **2**: 227-242.

Examples

```
# Processing Data with grouping variables.

titration$log_dil <- -log10(titration$dil)

SpearKarb(titration, positive + total ~ log_dil)
SpearKarb(titration, positive + total ~ log_dil | Operator)
SpearKarb(titration, positive + total ~ log_dil | Vial)
SpearKarb(titration, positive + total ~ log_dil | Operator + Vial)

skrmdb.all(titration, positive + total ~ log_dil | Operator + Vial)

## Monotonically increasing data

# The three calls are equivalent.
dead <- c(0, 3, 5, 8, 10, 10)
total <- rep(10, 6)
dil <- 1:6

## Use numeric vectors in the formula
DragBehr(dead + total ~ dil)
ReedMuench(dead + total ~ dil)
SpearKarb(dead + total ~ dil)
skrmdb.all(dead + total ~ dil)

data <- data.frame(y = dead,
                   n = total,
                   x = dil)

## Use data plus the formula
DragBehr(data, y + n ~ x)
ReedMuench(data, y + n ~ x)
SpearKarb(data, y + n ~ x)
skrmdb.all(data, y + n ~ x)

## Using y, n, x is deprecated.
DragBehr(y = dead, n = total, x = dil) |> suppressWarnings()
ReedMuench(y = dead, n = total, x = dil) |> suppressWarnings()
```

```

SpearKarb(y = dead, n = total, x = dil) |> suppressWarnings()

# The function automatically reorders the data as needed to
# meet the ordering assumption.
dead <- rev(dead)
total <- rev(total)
dil <- rev(dil)
SpearKarb(dead + total ~ dil)

## Unordered data
# Observe that the data is not monotonic after being sorted by dil.
dead <- c(10, 8, 5, 3, 0)
total <- rep(10, 5)
dil <- c(1, 3, 2, 4, 5)
SpearKarb(dead + total ~ dil)

```

print.skrmdb

print.skrmdb

Description

A print generic for a `skrmdb` object

Usage

```

## S3 method for class 'skrmdb'
print(x, rnd = 3, verbose = TRUE, ...)

```

Arguments

<code>x</code>	An object of class <code>skrmdb</code> .
<code>rnd</code>	The number of digits to round the numeric values.
<code>verbose</code>	If <code>TRUE</code> , prints out messages to help interpret the last five columns of the printed table.
<code>...</code>	Additional parameters.

Value

The results table in `x`, invisibly.

`skrmdb-class`*skrmdb class*

Description

The functions `ReedMuench()`, `SpearKarb()`, `DragBehr()`, and `skrmdb.all()` return an S3 object of class `skrmdb`.

The `skrmdb` class object contains the following fields:

- `eval`: The name of the method used to compute the ED50.
- `data`: A `data.frame` of the data used to compute the ED50.
- `ed`: A vector of ED50 values.
- `var`: A vector of variances, only if `SpearKarb` was used.
- `results`: A `data.table` of the results, including brief information about the data.

These components can be accessed via the appropriate accessor functions.

Usage

```
getED50(x)
```

```
getVar(x)
```

```
getData(x)
```

```
getResults(x)
```

Arguments

`x` An object of class `skrmdb`.

Examples

```
y <- c(0, 3, 5, 8, 10, 10)
n <- rep(10, 6)
x <- 1:6
res <- SpearKarb(y + n ~ x)
print(res)
getED50(res)
getVar(res)
getData(res)
getResults(res)
```

titration

Example Titration Data

Description

A data set that illustrates the dichotomous data format for titration data as found in the CVB Data Guide. The data are intentionally incomplete and nonmonotonic.

Usage

titration

Format

A data.frame with 67 rows and 9 columns.

- testID: Mandatory. A test identifier that is unique within table.
- PrepID: Mandatory. The identifier for the preparation used. This will usually be a vaccine lot or serial number.
- PrepRole: Mandatory. The role of the preparation. This must be "reference", "test", or "other".
- Date: Optional. The date the test was performed.
- Vial: Optional. The vial number tested.
- Operator: Optional. The operator who performed the test.
- dil: Mandatory. The dilution used in a well.
- positive: Mandatory. The total number of positive readings (tubes or wells) affected by the challenge.
- total: Mandatory. The total number of tubes or wells in a group at the specified dilution.

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